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Re: New Manuscript Submission, "Learning and the Efficiency of Physician Referrals."

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Dear Editors,

Enclosed please find our manuscript, "Learning and the Efficiency of Physician Referrals," to be considered for publication in the *Quarterly Journal of Economics*. We study how well referring physicians learn about specialist quality and act on that knowledge in their referral decisions, using data on 4.5 million major joint replacement surgeries from the universe of Medicare fee-for-service claims. We document substantial within-market heterogeneity in specialist quality, provide design-based evidence that primary care physicians adjust their referrals in response to their own patients' outcomes, and estimate a structural learning model to quantify the losses from informational frictions and the scope for reallocation under improved information.

The paper speaks to a question of broad interest well beyond health care, namely how expert intermediaries acquire and act on quality information, and how the resulting frictions shape market allocations. Incomplete information about quality and reliance on intermediating experts are common features of markets for complex goods, from professional and financial services to health care. Our setting provides an unusually clean opportunity to measure learning and its consequences, because we observe the full history of referrals, the quality-relevant outcomes that drive updating, and the choice set available to each referring physician.

The analysis relies on restricted-use data that we are not permitted to redistribute. The Medicare fee-for-service claims, Beneficiary Summary Files, and provider practice data are used under a Data Use Agreement with the Centers for Medicare and Medicaid Services (CMS) and are accessed within the CMS Virtual Research Data Center; the American Hospital Association Annual Survey is proprietary and available by license. Upon acceptance, we will deposit all analysis code and the permissible public data inputs to the QJE Dataverse with documentation sufficient for replication. The analysis is fully reproducible for any researcher who has obtained the CMS data through the standard application process and licensed the American Hospital Association data.

We confirm that this manuscript is not under consideration for publication elsewhere. Thank you for your consideration, and we look forward to receiving your comments and suggestions.

Sincerely,

Ian McCarthy